



Question Set

Question 1

Consider SOFR futures, OIS swaps, and collateralized USD swaps. Define the state variables (overnight rates, accrual factors, discount factors, futures convexity, and swap annuities) and derive the OIS discounting and par swap-rate equation. State the measure, filtration, and tradability assumptions.

$$K_{\text{swap}} = \frac{1 - P(0, T_n)}{\sum_{i=1}^n \alpha_i P(0, T_i)}$$

Question 2

For SOFR futures, OIS swaps, and collateralized USD swaps, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in SOFR futures, OIS swaps, and collateralized USD swaps. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using monthly and quarterly SOFR quotes used to bootstrap discount factors up to five years, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$K_{\text{swap}} = \frac{1 - P(0, T_n)}{\sum_{i=1}^n \alpha_i P(0, T_i)}$$

Question 5

For SOFR futures, OIS swaps, and collateralized USD swaps, compute and interpret bucketed curve DV01 and Jacobian-based quote sensitivities. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes SOFR futures prices and OIS par swap rates. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for SOFR futures, OIS swaps, and collateralized USD swaps. Explain how discount-factor positivity and monotone time-value limits prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for SOFR futures, OIS swaps, and collateralized USD swaps. Specify parallel, slope, and curvature shocks to the SOFR curve, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements SOFR futures, OIS swaps, and collateralized USD swaps and gets a stable but wrong answer. Explain how mixing simple, compounded, and continuously compounded conventions creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for SOFR futures, OIS swaps, and collateralized USD swaps. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of SOFR futures, OIS swaps, and collateralized USD swaps, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of monthly and quarterly SOFR quotes used to bootstrap discount factors up to five years. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for SOFR futures, OIS swaps, and collateralized USD swaps. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in SOFR futures, OIS swaps, and collateralized USD swaps. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for SOFR futures, OIS swaps, and collateralized USD swaps is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for SOFR futures prices and OIS par swap rates are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

$$K_{\text{swap}} = \frac{1 - P(0, T_n)}{\sum_{i=1}^n \alpha_i P(0, T_i)}$$

Question 17

For SOFR futures, OIS swaps, and collateralized USD swaps, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for parallel, slope, and curvature shocks to the SOFR curve. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that mixing simple, compounded, and continuously compounded conventions could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

$$K_{\text{swap}} = \frac{\frac{1 - P(0, T_n)}{n}}{\sum_{i=1}^n \alpha_i P(0, T_i)}$$

Question 20

Explain SOFR futures, OIS swaps, and collateralized USD swaps and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.